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Swarm robotics: a state-of-the-art review

Hauptseminararbeit

Sophia Guzmán Artmann

Betreuer: M. Sc. Paul Robert Schulze

Verantwortlicher Hochschullehrer:

Prof. Dr. H.-M. Groß

Fachgebiet: Neuroinformatik und Kognitive Robotik

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.....*Sophia Guzmán Artmann*.....
Sophia Guzmán Artmann

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Acronyms

chap	Chapter
sec	Section
subsec	Subsection
fig	Figure
alg	Algorithm
eq	Equation
Init	Initiate
UAV	Unmanned Air Vehicle
FSM	Finite State Machine
SI	Swarm Intelligence
PSO	Particle Swarm Optimization
Q	Q-learning
SARSA	State-action-reward-state-action
IR	Infrared Communication
LED	Light Emitting Diode
AI	Artificial Intelligence

Chapter 1

Abstract

Swarm robotics is an intriguing subject that features robots working together in a coordinated manner to achieve a common goal. These robots are typically simple and similar in design and interact locally to navigate their surroundings. Due to its diverse areas of application, it is easy to lose sight of the possibilities at hand. That is where this state-of-the-art review comes into play: it provides an overview of existing technologies, lists examples of real-life swarms, and options for practical implementations. The goal is to give a general understanding and overview of this discipline by summarizing research until now - hopefully fascinating more people in this field and enabling the creation of more robotic swarms!

Chapter 2

Introduction

Teamwork makes the dream work. Such is the case for research and development teams as well as for robots. Swarm robotics is a discipline of robotics that is heavily based on this principle.

Now, one might ask why this topic belongs to the faculty for neuroinformatics and cognitive robotics, when it deals with distributed algorithms and communication paradigms. These, however, are the only areas this document is based on. The true magic arises only after combining these topics and fusing them with robots that perceive their environment - an interdisciplinary task that requires the resources of the faculty for neuroinformatics and cognitive robotics.

2.1 Outline of document

This work is organized as follows: In Chapter 3 an overview of some of the current technologies is presented. For this, a broad introduction to swarm robotics and a summary of existing technologies and interactions with other swarms is given. Chapters 4 and 5 present examples, with the former mentioned one focusing on simulations and the latter on real-life robot swarms. Chapter 6 presents practical implementation advice collected during research. Finally, Chapter 7 is a summary of the document.

Chapter 3

Basics

3.1 Definition

Robotics aims to solve problems by developing robots specific to that task. A swarm or school is a term that's often used to describe a large number of insects or fish working in a coordinated manner to achieve a common goal. These swarms usually consist of the same species of animals that can only use their bodies, appendages, and environment to communicate and work as a team to survive. The members of the swarm tend to be relatively small, without many outstanding capacities if isolated from their peers.

Swarm robotics is a combination of both of these ideas: **many relatively simple, similar robots working as a team to reach a common goal using only local interactions** as sources of information. In other words, a (quasi) homogeneous (Subsection (subsec) 3.1.2) collective of simple robots (subsec 3.1.1) working towards a shared goal based on local interactions (subsec 3.1.4). Because of this, swarm robotics is typically characterized as being scalable, flexible, and robust ([Sahin et al., 2007]).

Scalability refers to the ability to add or reduce the number of robots in the swarm. Regardless, there are a few limitations on the size discussed in subsec 3.1.3. While scalability refers to flexibility in size, **flexibility** refers to responsiveness towards changes in the environment. As local interactions only allow a partial view of the final goal, the swarm is forced to repeatedly adapt to new environments. Finally, **robustness** represents the ability of the swarm being able to compensate for environmental noises and disturbances or hardware and software errors and defects ([Sahin et al., 2007]).

The exact definition, however, can vary from author to author. Some do not see a

necessity in having similar robots ([Dorigo et al., 2013]). Others emphasize limited to no control ([Arnold et al., 2005]) or intervention ([Sahin et al., 2007]) from human operators. There are even varying definitions of whether the members of the swarm must be the same and whether local communication is necessary ([Dorigo et al., 2013]). In this research paper, the definition in the previous section will be used as it is closest to the role model of naturally occurring swarms.

3.1.1 Simplicity

In this case, simplicity refers to **soft- and hardware simplicity**. In other words, to use only the strictly necessary components and as few as possible. This is an important criterion to keep the cost in proportion to the size of the swarm. Another advantage of having as little hardware as possible is a lower amount of points of failure ([Sahin et al., 2007]).

Algorithmic simplicity can be hard to achieve. Though this strongly depends on the final objective of the swarm (Section (sec) 3.2), algorithms and even the control or input to the swarm can present a mathematical challenge (subsecs 3.5.3, 3.4.5).

3.1.2 Heterogen vs homogen swarms

Another way of looking at hardware simplicity is by having all robots be the same. Working with such a homogeneous swarm will simplify mass production and reduce costs. Furthermore, if there is no difference in the capabilities of every robot, then any robot can be replaced without any issues, further adding to the robustness of the system. Additionally, the programming of this swarm is simpler as no distinctions between specialized groups or individuals need to be made, simplifying the swarm further. As such, some sources describe this as an essential feature in a robotic swarm ([Hamann, 2018], p. 5, 23) ([Abbass et al., 2018], p. 119). Most of the presented projects (Chapter (chap) 5) fulfil this criterion.

Some, however, argue that the system lacks flexibility. If all robots have the same capabilities, they all either have the most basic ones, limiting the entire swarm to its minimal usage, or they all have all peripherals needed for more complex tasks, raising the cost and complexity of the swarm. By introducing subgroups of robots with different peripherals, the versatility of the swarm is increased and therefore also the complexity

of the tasks the swarm can solve. Groups of robots can specialize in solving specific tasks, increasing the efficiency of the swarm ([Hamann, 2018], p. 23f, [Dorigo et al., 2013], p. 61). A swarm with these attributes is called "heterogeneous". The Swarmanoid project is a great example (subsec 5.4).

As a case can be made for both hetero- and homogeneous swarms, it would be best to find a compromise that includes the advantages of both concepts. In the same way a bee swarm works, a **"Quasi homogeneous swarm"** combines both the homogeneous and heterogeneous concepts. Most individuals are the same like worker bees, with only a few specialized robots like the queen or drones. Even though robustness through redundancy is reduced, they allow the swarm more flexibility by adding additional capabilities to the swarm.

3.1.3 Size

The number of robots considered too little or too many depends on the task, the complexity of the robots, and the size and intricacy of the environment. Generally, a minimum of three robots is required to be considered a swarm. As for the maximum, it is argued that the swarm should not function as a stochastic model, but rather as a distributed robotics model. Therefore the Avogadro constant, that is $6 \cdot 10^6$ individuals, was chosen as the maximal amount of individuals in a robotic swarm ([Hamann, 2018], p. 4, 9).

This, however, is a theoretical maximum. The practical maximum swarm size should be defined by peak performance, ensuring individuals do not block one another. For instance, in the "Stick Pulling Experiment" by [Ijspeert et al., 2001]. Here, the maximum size of the swarm is 11.

In this experiment, two robots must pull the stick out of the ground. To achieve this, each individual chooses a stick and waits for a helper to arrive. Should this not happen in a predefined amount of time, the robot will move to the next stick. The result of both the real-life swarm and simulation in webots (subsec 4.1.2) showed a higher performance if the swarm grows in a more than linear way per added robot for a swarm size of up to 6 robots, with the peak performance being 11 robots per swarm. After this, the robots blocked each other's attempts at pulling sticks ([Hamann, 2018], p. 4, 77). Of course, one could argue that increasing the size of the field or the amount

of sticks would also increase the size. According to the probabilistic model proposed by the inventors of the stick-pulling experiment, the size of the field scales proportionally to the number of robots for peak performance ([Ijspeert et al., 2001], p. 2).

3.1.4 Centralized vs Decentralized Communication

The communication between robots is also a size-limiting factor, as too many interfering signals might disturb each other. Currently, there are two main strategies of communication in the swarm: centralized and decentralized.

Centralized communication refers to the direct transmission of messages or orders to and from every robot. The user (human or computer) has a global view of the problem, allowing it to find the optimal solution by recognizing patterns and connections within the big picture. Using this information, it can tell every robot precisely what it should do to reach that state. After the problem has been identified, all robots can quickly finish their specific task, thus solving problems faster on average. The downside of this approach is that the problem has to be identified correctly and in a finite amount of time. Should this not be the case or the environment have changed after the initial recognition of the problem, the entire task has to be recalculated and all the instructions sent anew. Not having these messages interfere with each other and sent without an error can also present a challenge. Additionally, it is a battery-consuming process, not to mention the delay in time for each robot to send and receive messages from a central computer ([Hamann, 2018], p. 6, 86, [Dorigo et al., 2013]).

In contrast, **decentralized communication** is based on local interactions, that is when a robot can interact only with other neighbouring robots within the swarm. By limiting the individuals' range only a fraction of the whole problem can be seen at any given moment, provided they do not save their progress and update each other accordingly. Usually, this leads to a less optimal solution in a longer amount of time. Still, it allows the swarm to be more flexible, robust and scalable, as reactions to a changing environment are instantaneous. There is not one perfect solution to a problem, but rather a momentary best-effort result ([Hamann, 2018], p. 6, [Dorigo et al., 2013]). Hence, this approach is widely favoured in swarm robotics. Since this method relies heavily on **local interactions**, these terms will henceforth be used interchangeably. Security-wise, local interactions prove to be advantageous, too. By not connecting to

any wireless network not created by the swarm itself, the possibilities of a malicious user corrupting the swarm are limited to physical manipulation of the whole system, with no guarantee that this intervention will lead to the desired emergent behaviour (subsec 3.5).

Interestingly, pure decentralized communication is hardly possible due to the user having to give the swarm instructions on what their task is (subsec 3.4.5).

3.1.5 Types of communication

If information is only exchanged locally, no part of the swarm will know the entirety of the problem. Messages will get sent redundantly or their content might be outdated. Each robot will must keep track of all previous information and update the data and behaviour accordingly through explicit information exchange through direct means of communication such as radio waves, visual or auditory cues, or touch (subsec 6.2)([Ahmed and Glasgow, 2012]).

Another option is to outsource part of the data to the environment that, in some cases like collective Construction (subsec 3.2.1) or collective manipulation (subsec 3.2.3), will be changed by the swarm's actions at any rate. The so-called "**Implicit communication**" can also be achieved in the form of pheromone trails just like bees do ([Hamann, 2018], [Ahmed and Glasgow, 2012]).

Though it is a powerful tool, implicit communication makes less sense in situations with central communication or where the environment is less important, as is the case in sec 4 simulation platforms.

3.2 Areas of application

As for every new technology, projects in swarm robotics are widespread. Back in 2018 future potentials were seen in search and rescue missions, cleaning up oceans and exploring the galaxy ([Hamann, 2018], p. 28). As of 2023, most of these ideas have already been worked on in projects: areas struck by natural disasters are monitored by robotic swarms ([Blais and Akhlouf, 2023]), our oceans are being freed of plastic and oil by robotic swarms ([So and Lee, 2019], [Zahugi et al., 2013]), and mars is explored and spacecrafts repaired by robotic swarms ([Petrovsky et al., 2022], [Ebert, 2022]).

However, these are by far not the only areas of application of swarm robotics. Wherever a large area needs to be covered in a short amount of time such as in search and rescue (subsec 3.2.2), logistics ([Blais and Akhlouf, 2023]), applications in agriculture and forestry ([Blais and Akhlouf, 2023]), exploration missions on or off our planet ([Hamann, 2018], p. 27f), other types of environmental clean-up ([Debie et al., 2023]), site inspection ([Ebert, 2022], [Blais and Akhlouf, 2023]), object construction (subsec 3.2.1), or lead swarms of animals (sec 3.4.3), there are projects based on swarm robotics ([Blais and Akhlouf, 2023]). In short, whenever a task is monotonous and parallelization is possible, swarm robotics is the key (subsec 3.4.2).

Usually, projects specialize on one or two aspects of the following list:

- Collective construction (subsec 3.2.1)
- Search and Rescue (subsec 3.2.2)
- Collective manipulation and transportation (subsec 3.2.3)
- Flocking and collective motion (subsec 3.2.4)

3.2.1 Collective construction

Commonly seen in ant or termite colonies, collective construction refers to a swarm of robots building objects either from themselves or other objects. Not only do the algorithms must break down a body into simple instructions, but also the hardware design of the robot has to be capable of fulfilling all these tasks with a certain degree of accuracy. ([Hamann, 2018], p. 75ff). So far, there are only a few projects that have been successful at building objects, let alone 3-dimensional constructs. One of them is TERMES.

TERMES (subsec 5.2) uses both unmovable (passive) blocks and active (movable) robots. To accomplish their task, a specific set of local rules for each robot must be created based on the final result while keeping in mind that all areas must be reachable throughout ([Werfel and Nagpal, 2018]).

The preprocessing algorithm begins with a "seed". Starting from there, blocks can only be placed on top or next to each other. Once a block is attached, it can not be removed. Apart from this rule there are two more requirements. Firstly, when building in 3D, there can not be any holes in the structure, as it may collapse. Consequently, just like in 3D printing, a support has to be included underneath. Secondly, when dealing with

narrow passages, the construction must ensure that a robot and a building block can pass ([Werfel and Nagpal, 2018]).

Based on these rules, the preprocessing stage decides where blocks can be placed and under which condition. This information is then sent to all robots, who, while moving, analyse where they should go based on the already placed building blocks. Within this structure, the robots can move in any direction, carry passive building blocks, and avoid collisions. The robots follow a set loop: get block, move along the surface to place block, attach block ([Werfel and Nagpal, 2018]). They only communicate implicitly (subsec 3.1.5). Figure (fig) 3.1 demonstrates this process.

Blocks, in contrast, start communicating once placed on the structure. Information on attached blocks and invalid locations is exchanged through touch (subsec 6.2.1). Through this network, the attachment of blocks is allowed or prevented through communication of each block's state. Depending on the size of the structure and the number of members of the swarm, the communication can be rather costly ([Werfel and Nagpal, 2018]). The optimal size of the swarm should therefore be found before deployment (subsec 3.1.3). As a whole, this algorithm manages to operate a swarm based only on one initial input from a user and succeeds in preventing emergent behaviour (subsec 3.5.3).

Another example of a swarm that can build real-life constructs is the Kilobots (subsec 5.1). These coin-sized robots can create a picture-like plane by placing themselves in a specific configuration, as seen in fig 3.2.

3.2.2 Search and rescue (foraging and exploration)

As teased in subsec 3.2.3 collective manipulation and transport, a heterogeneous group of robots is the best example for this category. Swarmanoid (subsec 5.4) consists of 3 - 5 different types of robots. Individually, they excel at their individual task, but when working as a group they can autonomously rescue books, as can be seen in the following video <https://www.youtube.com/watch?v=M2nn1X9Xlps> or in fig 3.3. Pictures of the individual robots can be seen in fig 5.4 and fig 5.3. Swarmanoid operates in the following way:

1. The eye-bot finds the desired object.

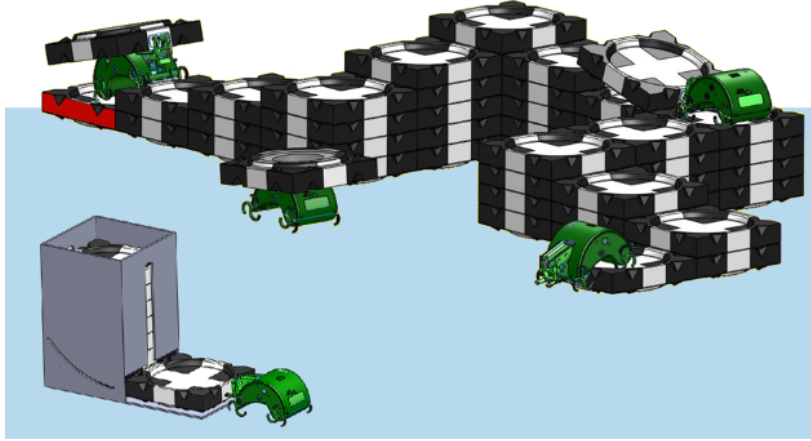


Figure 3.1: TERMES in progress

The individuals follow the algorithmic cycle as discussed above: Upper left robot: got a block, following the path until an allowed position is reached. Upper right robot: Got block, got the location, attaching block. Lower right robot: Got block, got location, attached block, currently getting a new one. Lower left robot: collecting new block. Last robot: Got block, restarting process. The seed is the red building block ([Petersen et al., 2008], [Werfel and Nagpal, 2018]).

2. The foot-bots (also called swarmbots subsec 5.3) then form a path to the object from the swarm's starting location.
3. Meanwhile, two foot-bots pick up a hand-bot and transport it to the desired location. Simultaneously, transportation of a second hand-bot in case of several objects or failure of the previous envoy begins.
4. Once the hand- and foot-bot-envoy has arrived at the destination, the hand-bot retrieves the desired object.
5. Finally, the hand-bot with the book and the foot-bots return to their home base to await further instructions from the eye-bots.

3.2.3 Collective manipulation and transportation

Collective Manipulation refers to several robots working together to move or change an object. An example of a robotic swarm lifting objects is the stick-pulling experiment

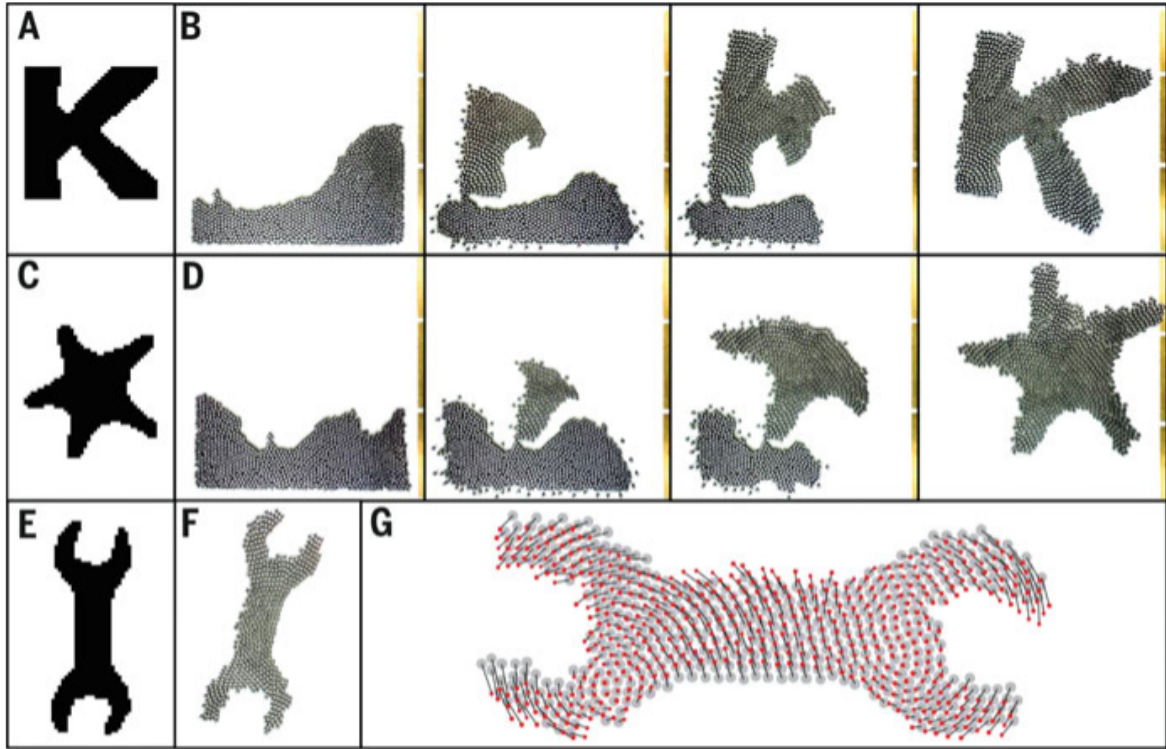


Figure 3.2: Kilobots in progress

From left to right: the first picture is the input image. The other pictures show the process of the swarm transforming into the input image until the last picture on each row is the final shape of the swarm. Picture G is a comparison of the accuracy of the location of the individual robots in comparison with the input image ([Hamann, 2018], p.73).

(subsec 3.1.3). In warehouses, however, picking up sticks is not enough. Items of varying shapes, weights and sizes must be transported safely from one location to another without collisions. For this task, the robots must first find the object, determine how many individuals are needed and then carry it to its desired location.

In the algorithm proposed by [Jurt et al., 2022], the robot moves randomly to find the desired object. The robot will now determine its position underneath the object by counting the number of passing robots and asking them about their current state. If a certain threshold of visiting robots with the state of "finding cluster" is detected, the robot will know that it's close to the border of the object. Should, in turn, a certain number of stationary robots be within a certain radius and waiting on a positive "safe-to-lift" response, it will know that it is fully underneath the object. Both of these

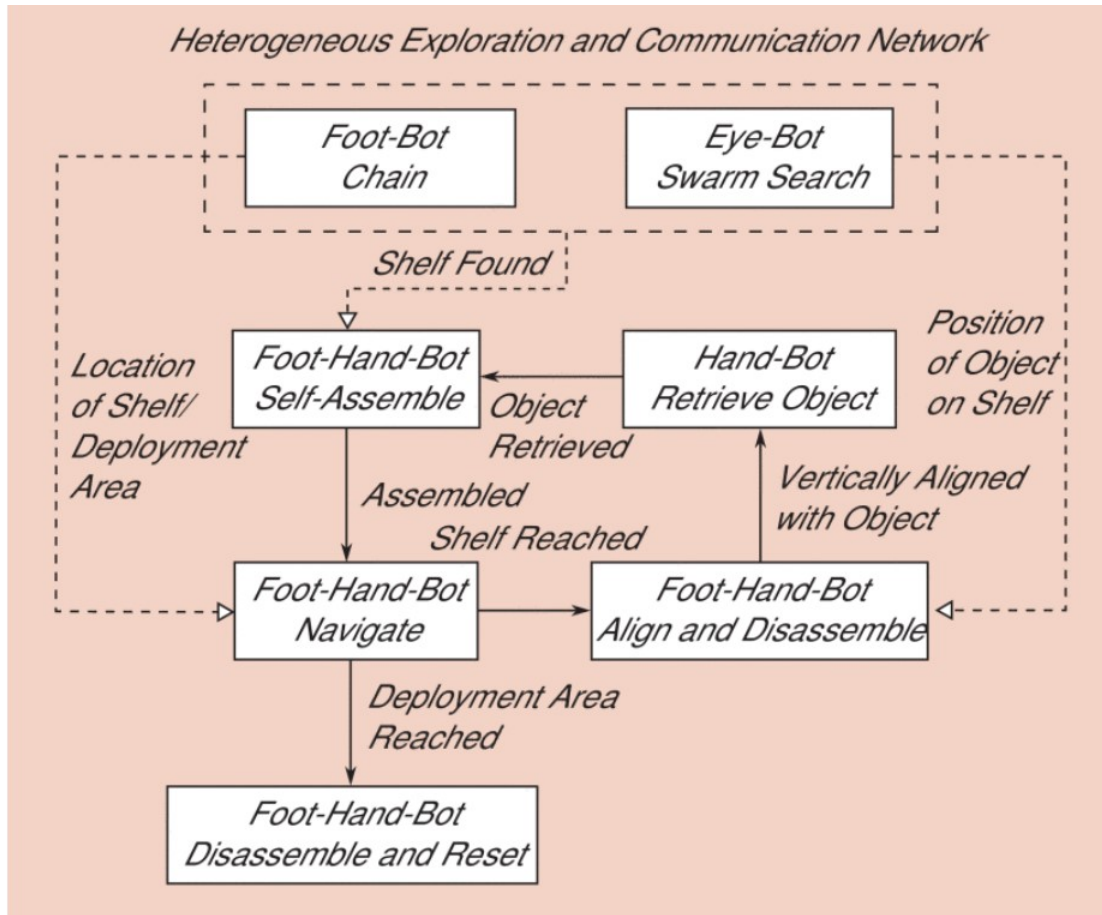


Figure 3.3: Flow diagram Swarmanoid

An alternative, more detailed description of the Swarmanoid algorithm as a flow diagram. The dotted lines represent wireless data exchange, and the solid lines show when another robot class takes up the task. Taken from [Dorigo et al., 2013]

states will be broadcast. Once all robots under the object are in a "ready-to-lift" state, the object will be lifted and moved randomly until the user sends a stop signal. The swarm will proceed to find a new object to carry. There is a flow-state-diagram in fig 3.4 and a pseudocode in Algorithm (alg) 3.1 to make the algorithm more tangible.

However, the algorithm has a few limitations. For instance, it operates under the assumption that weight is uniformly distributed. The probability of success within a finite time decreases as the size of the object increases. Objects larger than $3m^2$ can not be transported at all. Additionally, as the size of the swarm required to transport the object increases, the communication cost grows exponentially. Therefore, further

development is needed before this technology can be used in real-life warehouses ([Jurt et al., 2022]).

Another example of a swarm that can build real-life constructs is the Kilobots (subsec 5.1). These coin-sized robots can create pictures by placing themselves in a specific shape, creating a picture-like plane as seen in fig 3.2.

Algorithm 3.1: Pseudocode Collective Transport and Motion

```

1 Init robot state  $s$  = search object
2 Init robot number  $i$ 
3 Init counter for passing robots with the state "find cluster"  $c_{fc} = 0$ 
4 Init counter for passing robots with the state "safe-to-lift"  $c_{stl} = 0$ 
5 Init threshold for state "findCluster"  $c_{fcmax}$ 
6 Init threshold for state "safe-to-lift"  $c_{stlmax}$ 
7 while 1 do
8    $randomMove()$ 
9   if  $cluster\ detected$  then
10     $s \leftarrow positioned$ 
11    while  $s == positioned$  do
12       $broadcastState()$ 
13      if  $s_{passingRobot} == s_{findCluster}$  then
14         $c_{fc} \leftarrow c_{fc} + 1$ 
15      if  $s_{passingRobot} == s_{safe-to-lift}$  then
16         $c_{stl} \leftarrow c_{stl} + 1$ 
17      if  $c_{fc} == c_{fcmax}$  then
18         $s \leftarrow findCluster$ 
19      if  $c_{stl} == c_{stlmax}$  then
20         $s \leftarrow safe-to-lift$ 
21    if  $s == safe-to-lift \ \&\ s_{allSurroundingRobots} == safe-to-lift$  then
22       $s \leftarrow moveObject$ 
23    while  $!stopMessage$  do
24       $randomMove()$ 
25     $s \leftarrow leaveObject$ 
26    while  $objectSensor$  do
27       $randomSensor()$ 
28     $s \leftarrow searchObject$ 

```

In search and rescue missions (subsec 3.2.2), objects must be transported and stored as well. For this, the Swarmanoid team (subsec 5.4) received new members: book-bots (an object in a book-like shape that has to be transported) and cart-bots (a robot designed

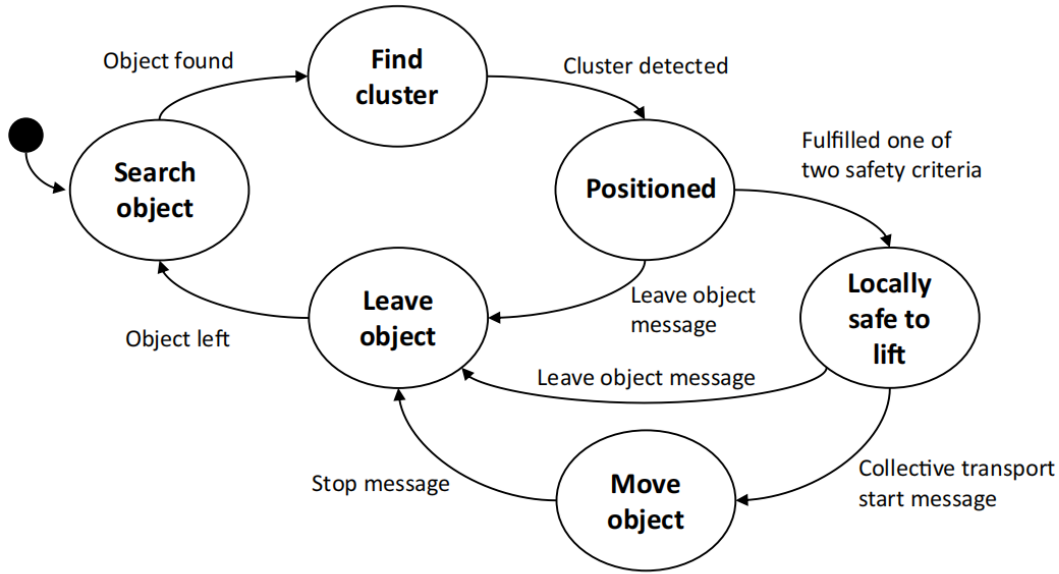


Figure 3.4: FSM Collective Transport

Algorithm used by [Jurt et al., 2022] to reliably transport objects of arbitrary size, shape, and weight.

to store the book-bots) ([Decugnière et al., 2008]).

3.2.4 Flocking and collective motion

Heavily based on natural pattern occurrence in nature and human behaviour, algorithms of the "pattern formation" category usually involve leaving virtual pheromones and trail networks ([Hamann, 2018], p. 69).

A different approach would be to use reinforcement learning algorithms. To plan a path and navigate to the desired location, a mixture of the Q-learning (Q) and Particle Swarm Optimization (PSO) (subsec 3.4.2) algorithms is employed. Then, adding a deep-State-action-reward-state-action (SARSA) allows the robot to reach its destination without any collisions ([Blais and Akhlouf, 2023]).

3.3 Prerequisites

In order to be able to achieve success in one of the previously mentioned fields, there are basic functionalities that are almost always required.

3.3.1 Sorting

Sorting occurs naturally, and algorithms can use it by mimicking nature's methods. For instance, the "Brazil nut effect" is a naturally occurring sorting mechanism. Here, nuts determine their height in a jar based on their size. In other words, clusters automatically form inside the swarm based on a specific attribute ([Hamann, 2018], p. 70f).

3.3.2 Aggregation and Dispersion

Aggregation and dispersion refers to the ability of a group of robots to either accumulate or separate and explore on call. This is important, as the environment is always changing and so is the optimal behaviour of the swarm. If required, the swarm must find new options but always keep comparing them to previous solutions.

The biologically-inspired variant to achieve this is by emulating pheromones. This method is called positive and negative feedback as it either encourages or discourages the exploration and expansion of the swarm as opposed to having decided to have found the optimal solution to a feature([Hamann, 2018], p. 66, 68).

Where that optimal solution is depends on the environment and presents its own challenges and will be further discussed in subsec 3.3.3. Usually, a spot is found that is optimal based on a certain criteria.

3.3.3 Collective decision-making

When it comes to collective decision-making in a swarm, there are a few approaches that can be taken. One option is to use the leader-follower-principle, where a leader is either chosen through reinforcement learning algorithms such as SARSA or Q (subsec 3.2.4)([Blais and Akhlouf, 2023]), p. 232) or randomly selected by the swarm or the user.

Another option is to model a decision as a convergence towards a state. In this approach, all the members of the swarm must reach the same state in a finite amount of time. The latter approach is often modelled as a majority vote. Here, all robots adapt to the predominant state of their immediate neighbours. As such, when faced with binary decisions, it leads to random results. Others have therefore adapted this model to either move towards the average (Hegelsmann und Krause), synchronize the swarm

using wave functions (Kuramoto and variations), or update each individual's spreading function based on the innovation and exploration approach (Bass Diffusion Model) ([Hamann, 2018], p. 144-152).

Ultimately, these are just models. Many do not consider changes in the environment and get stuck on one option. Other models do not even converge towards an option. To combat this, either a noise factor can be added (Hegelsmann und Krause, [Hamann, 2018], p.145f) or so-called "anti-agents" can be included. These are agents that will always oppose the majoritarian opinion to avoid consensus and thus encourage the swarm to explore new options (subsec 3.3.2 and adapt to new environments (Sociophysics-and-Contrarians model, [Hamann, 2018], p. 152ff).

Like the previously mentioned models, the "Bayesian Decision-Making Algorithm" (BDMA) is an example of a practical implementation of a variation of the majority model. It combines that model with Bernoulli- and Beta-Distribution functions to reach consensus in a finite amount of time. In this specific example, each robot estimates the amount of white in a binary environment in the form of a Beta function. While moving, each robot senses the environment, broadcasts that information, and updates its own colour according to its decision. This algorithm proposed by [Ebert, 2022] was designed to determine whether the environment is more black or white. A pseudocode for this can be found in alg 3.2. It follows the following procedure:

1. Lines 1-5: At the start, the robots are placed randomly in the desired area. All variables are initiated.
2. Lines 6-12: Each robot then moves towards the edge of the area with the desired quality, saving an observation C every few τ seconds.
3. Lines 13-16: The robot's own observation is updated according to the new information and the incoming messages m .
4. Lines 19-22: A decision is made on whether the individual's personal criteria are met.
5. Lines 23-26: The observation is broadcast to neighbouring robots and the loop restarts.

As this algorithm can only decide on one characteristic, an updated version of it, the Multi-Feature Decision-Making Algorithm (MFDM), has been developed. The difference mainly consists in additionally considering the concentration of the robots,

Algorithm 3.2: Pseudo-code Bayesian Decision-Making Algorithm

Input : Observational interval τ
 credible threshold p_c
 prior parameter α_0
 positive feedback indicator u^+
 robot UID id

Output : Binary classification of environment d_f

```

1  Init counter white observation  $\alpha = \alpha_0$ 
2  Init counter black observation  $\beta = \alpha_0$ 
3  Init observation index  $i$ 
4  Init incomplete decision  $d_f = -1$ 
5  Init dictionary of received observations  $s = ID : (0, 0)$ 
6  for  $t \in [1, T]$  do
7      Perform pseudo-random walk
8      if  $\tau$  divides  $t$  then
9           $C \leftarrow$  observed colour (0, 1)
10          $\alpha \leftarrow \alpha + C$ 
11          $\beta \leftarrow \beta + (1 - C)$ 
12          $i \leftarrow i + 1$ 
13     Let  $m = (id', i', C')$ 
14     if  $s(id') \neq m(id')$  then
15          $\alpha \leftarrow \alpha + C$ 
16          $\beta \leftarrow \beta + (1 - C')$ 
17     if  $d_f = -1$  then
18         Let  $p$  denote the cumulative distribution function of Beta ( $\alpha + \alpha_0, \beta + \alpha_0$ ) at
19         0.5
20         if  $p > p_c$  then
21              $d_f \leftarrow 0$ 
22         else if  $(1 - p) > p_c$  then
23              $d_f \leftarrow 1$ 
24         if  $(d \neq -1) \ \& \ u^+$  then
25             Broadcast message  $(id, i, d_f)$ 
26     else
27         Broadcast message  $(id, i, C)$ 
```

influencing the belief, and thus allowing it to evaluate multiple features simultaneously. This algorithm has proven to be finite and accurate. Interestingly, both the BDMA and MFDm perform significantly better in (quasi)homogeneous systems and when the options are fundamentally different ([Ebert, 2022], p.52f).

Still, these algorithms are still relatively recent. As such, in practice, it is more common to see other algorithms like BEECLUST. Inspired by honeybees, this algorithm aims to decide whether to explore the environment or stay at the optimal point - in short, solving the aggregation and dispersion problem (subsec 3.3.2). The robot will remain at the possible best location for a time proportional to the quality coded as the intensity of the desired trait ([Schmickl and Hamann, 2011]).

To achieve the desired behaviour, the adapted algorithm in alg 3.3 can be used, or the following steps can be taken ([Schmickl and Hamann, 2011]):

1. Each robot moves as straight as it can until it encounters an obstacle in front.
2. If this obstacle is a wall (or any other non-robot barrier), it turns away and continues with step 1.
3. If the obstacle is another robot, the robot measures the local luminescence. The quality of the location is calculated based on the minimum value between the measurement and a lower threshold.
4. The more intense the desired trait is, the longer the robot stays immobile in the place. The waiting time is calculated through the formula $w_i = \max(1, \frac{w_{max} \cdot e_i^2}{e_i^2 + 7000})$ whereby w_{max} is the maximum waiting time, e_i the calculated value for the intensity, and i is the number of the robot.
5. After the waiting time is up, the robot turns away from the other robot and proceeds with step 1.

A video showing the resulting behaviour is available at <https://www.youtube.com/watch?v=Q1NBreg13hM>. However, it can be easily intercepted through a defective robot or an individual purposefully hindering the process. Although biasing has been tested as a solution, it is still recommended to introduce an additional control mechanism to prevent disruptions ([Schmickl and Hamann, 2011]).

Algorithm 3.3: Pseudo-code BEECLUST algorithm

```

1 Init generic counter  $c = 0$ 
2 Init max counter  $c_{max}$ 
3 Init intensity  $e_i$ 
4 Init max waiting time  $w_{max}$ 
5 Init robot number  $i$ 
6 Init lower threshold  $e_{min}$ 
7 Init measurement of quality  $q$ 
8 while  $c \geq c_{max}$  do
9    $w_i = 1$ 
10   $moveStraight()$ 
11  if  $obstacle \neq robot$  then
12     $turn()$ 
13  else
14     $e_i \leftarrow \max(e_{min}, q)$ 
15     $w_i \leftarrow \max(1, \frac{w_{max} \cdot e_i^2}{e_i^2 + 7000})$ 
16     $wait(w_1)$ 
17   $c \leftarrow c + w_i$ 

```

3.3.4 Optimization

Still, many solutions found so far are specific to that problem. In real life, the swarm must adapt to changing environments and other issues. As such, [Fister et al., 2013] has developed an algorithm to find the optimal solution in every situation. It analyses the available solutions and picks the one most suited for the new environment. Though originally meant as Swarm Intelligence (SI) (subsec 3.4.2) algorithm, it was adapted for use in robotic swarms.

The firefly algorithm was developed for homogeneous swarms and relies on stochastic principles. In the adapted algorithm in alg 3.4, the attractiveness is proportional to the light intensity I , described by $I(r) = I_0 \cdot \exp(-\gamma \cdot r^2)$ and based on a semi-periodic wave function. I_0 refers to the light intensity of the source. r is calculated through the Euclidean distances between two robots. Regarding the attractiveness of the firefly β , it can be computed using $\beta = \beta_0 \cdot \exp(-\gamma \cdot r^2)$ with β_0 representing the attractiveness at the start. γ usually ranges between 0.1 and 1. α is used to calculate the best solution s^* . Lastly, *MAXREP* is the maximum number of repetitions according to the individual fitness function ([Fister et al., 2013]).

Algorithm 3.4: Pseudo-code Firefly algorithm

Output : Best solution s^*

```

1   Init generic counter  $t = 0$ 
2   Init best solution  $s^*$ 
3   Init absorption coefficient  $\gamma = 1.0$ 
4   Init attractiveness  $\beta$ 
5   Randomly init population  $P^{(t+1)}$ 
6   while ( $t < MAXREP$ ) do
7       Determine a new value for  $\alpha$ 
8       Evaluate  $s$  according to  $f(s)$ 
9       Sort  $s$  according to  $f(s)$ 
10      Determine the best solution  $s^*$ 
11      Vary the attractiveness and move the individual to the position  $P^{(t+1)}$ 
12       $t = t + 1$ 

```

3.4 Swarm robotics and other technologies, insects, and humans

After having analyzed how robots within a swarm interact with each other, it might be interesting to see how the same robots interact with animals, humans, or competing technologies.

3.4.1 Swarm robotics and individual robots

Teamwork is the principle of robotics swarms. Therefore, robots used in competitive team sports based on swarm robotic principles should beat the other team of individual players. Such was the case in a Robocup soccer team competition, where the swarm robotics team easily defeated the local champions. This win is credited to their team-mate-oriented approach using a heuristic implementation of SI algorithms ([Carrera, 2006]). In general, the core principles of robustness, scalability, and flexibility (sec 3.1) are advantageous in most situations. Still, a swarm robot will be useless when alone, while a robot designed to work individually might perform a task seamlessly. Additionally, the final monetary cost of the entire swarm may surpass the one of the individual robot.

Ultimately, every problem has many solutions, some of which might be more effective

than others. It is the developer's task to decide the best technology for their problem.

3.4.2 Swarm robotics vs Swarm intelligence

Swarm Intelligence (SI) is a field that focuses on developing algorithms that involve multiple actors based on biological role models. In contrast to swarm robotics, swarm intelligence is usually bound to a stationary computer, thus assuming unlimited computational capacity and battery, optimal environmental conditions, and infinite individuals are available. Still, these algorithms are used in different areas, for instance, power and voltage control, evolving weights in neural networks, route finding, and modelling the spreading of diseases in bodies ([Ahmed and Glasgow, 2012]). Usually, these algorithms can be modified for swarm robotics as seen in subsec 3.3.4.

As each discipline's areas of applications are very different from one another and swarm robotics draws inspiration from SI, these two disciplines are difficult to compare. An example of a SI algorithm that is also very often used in swarm robotics is the PSO algorithm. Based on pheromone trails of ants, this algorithm aims to control the flocks or swarm movement by assembling individuals, avoiding collisions, and matching their velocities. The algorithm works by sharing each individual's "fitness" (the quality of their solution), comparing, and adjusting to it. Here, not only the current highest fitness between robots is compared, but also the historically best fitness values are taken into consideration. PSO has been adapted for various applications, such as adjusting the biases of feed-forward multi-layer perceptron neural networks to control the movement of a robotic swarm. A basic pseudocode would look like alg 3.4.2.

The performance can be further improved by introducing a maximum velocity, having an inertia weight, and a neighbourhood topology. There are also versions of this algorithm that do the comparisons locally, enabling decentralized communication (subsec 3.1.4). According to ([Ahmed and Glasgow, 2012]), A basic version would work as follows:

1. Initialize the swarm by randomly assigning each particle to an arbitrary initial velocity and a position in each dimension of the solution space.
2. Evaluate the desired fitness function for the individual particle's optimal position.
3. For each particle, update its historically best position P_i if its current position is better.

4. Identify or update the swarm's globally best particle with the swarm's best fitness value, and set or reset its index g and its position P_g .
5. Update velocities of all the particles using Equation (eq) 3.1 equation.

$$v_{id}(t+1) = v_{id}(t) + c_1 \cdot R_1(p_{id}(t) - x_{id}(t)) + c_2 \cdot R_2(p_{gd}(t) - x_{id}(t)) \quad 3.1$$

whereas v_{id} is the velocity, x_{id} the position, p_{id} the historically best position, p_{gd} the global best position, R_1 and R_2 is a factor to introduce randomness, and c_1 and c_2 weighing parameters determining the importance of the particle's own experiences in comparison to the swarm's. The i refers to the number of particles, d to the dimension.

6. Move each particle to it's new position using eq 3.2

$$x_{id}(t+1) = x_{id}(t) + v_{id}(t+1) \quad 3.2$$

7. Repeat steps 2-6 until convergence or a stopping criterion (max number of allowed iterations/ sufficiently good fitness is achieved/ algorithm has not improved performance for several iterations) is met.

3.4.3 Swarm robots in a new light: interaction with insects and animals

Experiments have shown that influencing animal swarms' behaviour through swarm robotics is possible. The only requirement is a robot that targets their main sensory organ. For instance, a little robot called "InsBots" (subsec 5.5) can steer a swarm of cockroaches. It achieves this by coating the robots (see fig 3.5) with pheromones necessary for cockroach communication ([Caprari et al., 2005]). Other projects have achieved similar results with young chickens and swarms of bees ([Hamann, 2018] p. 87f).

As for a more recent example, researchers were able to influence the direction of a swarm or shoal of zebrafish using a combination of visual cues, underwater motion, and weak electrical impulses. Unlike cockroaches, the fish are more reactive to visual cues than pheromones, a fact that can also be seen when comparing two versions of the RiBot (subsec 5.6) robots in fig 3.6 ([Bonnet et al., 2018]).

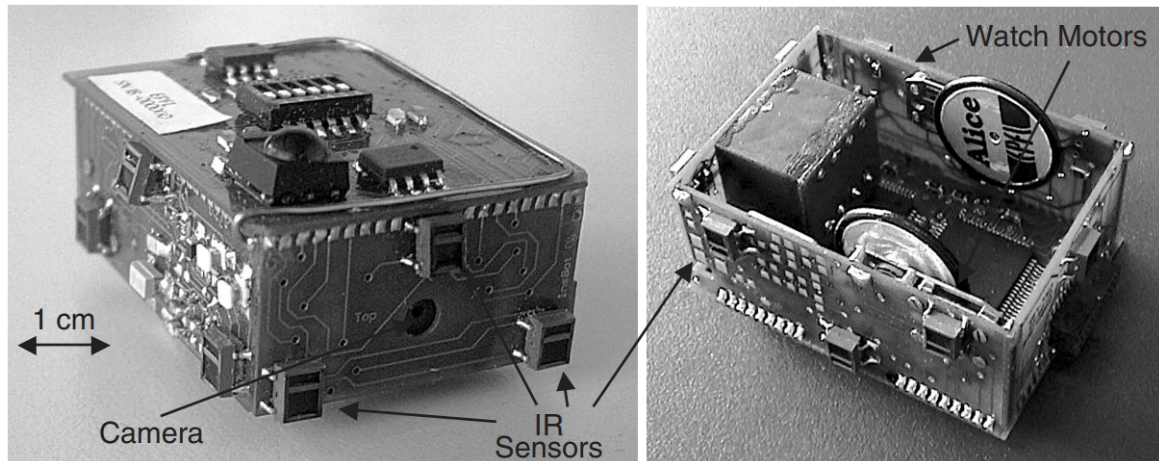


Figure 3.5: InsBot

Left: InsBot. Right: InsBot upside down; without battery nor bottom cover. Even though it does not look like a cockroach at all, it has succeeded in leading the swarm through its pheromones ([Caprari et al., 2005], p. 61).

3.4.4 Human perception of robotic swarms

Just like cockroaches, autonomous swarms of little robots are usually presented in a negative light in movies and other forms of media. Additionally, most swarm behaviours are modelled after bees (subsec 26) or other insects like termites (subsec 5.2), animals that many people have reservations about. This dislike is further amplified the bigger the swarm and the smaller the members get ([Hamann, 2018], p. 1, 23).

3.4.5 Designing control made for humans

Human reactions to the swarm aside, the swarm needs to know what it is supposed to do and if it is doing it correctly. For this, a common approach seems to be to make a grid in the desired dimensions and have a compiler make local rules for each robot considering heavy restrictions, similar to subsec 5.2 ([Kolling et al., 2016]).

For humans, it is easier to steer the swarm when only presented with the mean and the variance of the swarm's movement, as it significantly simplifies visual feedback ([Shahrokhi and Becker, 2015]). This way, only the global control signals are sent that later need to be distributed and organized so an individual can adapt to its movement, a task that presents a challenge on its own (subsec 3.5).



Figure 3.6: RiBot

Left: The version of the RiBot with a zebrafish colour pattern. Right: The RiBot version without any zebrafish colour pattern enclosure. The left version has proven to be more efficient ([Bonnet et al., 2018], p. 506).

Instead of broadcasting updates to all members of the swarm, information can be passed on locally through a single individual via algorithms, parameters, or environmental cues. This method is of particular use in areas where other disturbances might affect communication options or limit the bandwidth as is the case during search and rescue missions (subsec 3.2.2). This type of communication can suffer from delays and asynchronous spreading of information. However, the use of predictive algorithms can reduce performance issues of this origin ([Kolling et al., 2016]).

3.5 Emergence

As mentioned in subsec 3.4.5, a user would prefer to communicate with one swarm entity in contrast to every single robot separately. In other words, a user will want to send "global" or "macroscopic" instructions for a certain behaviour for the swarm as opposed to a "local" or "microscopic" order to a single robot. ([Hamann, 2018], p. 15). That instruction must be translated to an individual's movement in context to the entire swarm's movement using decentralized communication. Emergence describes the global

behaviour of the swarm that diverges from the microscopic behaviour ([Abbass et al., 2018], p. 117). Though an individual's behaviour might be accurately programmed in many situations, the swarm as a whole exhibits a very complex, difficult to predict behaviour (subsec 3.5.3) ([Hamann, 2018], p. 21f).

3.5.1 Advantages of emergence

This is exactly the behaviour that swarm robotics aims for, as the swarm is supposed to succeed in tasks that an individual robot cannot do alone. Because all robots compute for themselves, this makes tasks highly parallelizable and autonomous from external sources ([Abbass et al., 2018], p. 119f). This last point also increases the security from the swarm towards malicious wireless attacks.

3.5.2 Disadvantages of emergence

The downside of emergent behaviour is that control of the entire swarm is only indirectly possible through control of the individual agents, making it nearly impossible to accurately predict the swarm's behaviour. However, in some cases, it can be necessary for the project at hand. Unpredictable swarm behaviour can introduce security risks, reduced accuracy or other issues in practical use. It may even lead to the robotic application not receiving the permit to be distributed by the government ([Abbass et al., 2018], p.120f).

3.5.3 Predicting and implementing emergent movement

Predicting the emergent behaviour is difficult but necessary for projects. Therefore, an alternative would be to program the swarm's macroscopic behaviour and extract local orders and information from it, just as in subsec 5.2. Though it can lead to impressive results as can be seen in fig 3.1, in general, it is specific to each task, meaning the details must be reprogrammed every time anew (subsec 5.2, [Abbass et al., 2018]). Though there are programming languages such as "Proto", that allow the user to compose the rules, this task can be very resource-consuming ([Kolling et al., 2016]).

Still, taking into consideration that this phenomenon is essential to swarm robotics, models have been developed to reliably predict the swarm's behaviour. According to

[Abbass et al., 2018], mathematical models based on temporal logic and statistical graphs were successful in predicting emergent behaviour with a high degree of accuracy, albeit only in theory. Ultimately, the decision to implement an emergence-conscious approach depends on the project and its intended application and should be implemented from the beginning ([Abbass et al., 2018], p.122).

Chapter 4

Simulation Platforms

An implementation, however, is challenging as the existing software and programming languages were not developed for the specific requirements of swarm robotics. Initial efforts to address this issue propose introducing frameworks as add-ons to existing programming languages.

Developed specifically for heterogeneous swarms, BUZZ treats the swarm like an object but each individual as a separate device. Using BUZZ allows the user to fall back on previously written information and relative distance calculating functions. On top of that, a pre-programmed step for each robot automatically measures the input, processes it and calculates the appropriate physical reaction ([Pinciroli and Beltrame, 2016]).

Another add-on for existing languages developed for heterogeneous swarms and specialized in self-assembly is the SWARMORPH-script. Based on the local cell-clustering called "morphology", a set of control primitives was designed based on the Swarmbots (Subsection (subsec) 5.3) ([Christensen et al., 2008]).

More difficulties in implementation include unpredictable disturbances or environmental noise. Other factors that may prevent enough tests from being completed include limitations in battery life, budget, space, and real-time measuring and monitoring. Usually, a robot that fulfils the projects' hyper-specific requirements has to be developed on top of the software itself, making projects like these time-intensive and costly. In contrast, simulation platforms provide better speed, simplicity, and detailed logging, simplifying testing of the algorithms. However, by ignoring real-life difficulties, accuracy is lost ([Cianci et al., 2007]).

4.1 All-purpose simulation platforms

Some developers prefer using well-rounded, established simulation platforms. These are developed and tested by specialists in this field, which saves resources and probably have better quality. Some of the platforms are listed below.

4.1.1 ARGoS

One of the most popular simulation platforms for swarm robotics is ARGoS and was most notably used for the Swarmanoid ([Pinciroli et al., 2011]) and ePucks (subsec 5.7) projects. In the latter project, it was able to control up to 1000 robots at once. It specialises in featuring a variety of robots and sensors.

In comparison to other simulation platforms, ARGoS performs the fastest by efficiently distributing computations between the machines' cores ([Ebert, 2022]). As a result, the run-time increases linearly with the number of robots ([Pinciroli et al., 2012], p. 272). Though the robots and their environments can be precisely simulated and various communication plug-ins are available, simulations are limited to ground-operating robots. It is written in C++ but also supports libraries such as Tensorflow or Pytorch, making it a suitable testing environment for Artificial Intelligence (AI)-operated swarms ([Blais and Akhlouf, 2023]).

It can be downloaded for free on the website <https://www.argos-sim.info/>

4.1.2 Webots

Used for the stick-pulling experiment (subsec 3.1.3) and the ePucks (subsec 5.7), this all-purpose robotic simulator with a 3D-physics engine is known for its reliable performance. However, due to its high hardware requirements, it becomes slower the more robots are being simulated simultaneously ([Ebert, 2022]).

Webots can be downloaded free of charge from the website <https://cyberbotics.com/>

4.2 SI and swarm robotics specific platforms

As already mentioned, there are many difficulties when simulating for Swarm Intelligence (SI) and swarm robotics purposes. As such, some simulation platforms like MARS

(marine robotics swarms) or Übersim (multi-robot soccer simulator) have been created to specifically address those issues ([Ebert, 2022]). Other notable examples were predominantly developed to explore and enhance the Kilobot’s (subsec 5.1) capabilities.

4.2.1 Kilombo

Kilombo is the first open-source C simulator written and designed for the Kilobots (subsec 5.1). Its main objective is the simulation of the movements and the IR communication between the Kilobots realistically ([Gebhardt and Neumann, 2018]).

It can be downloaded through this website <https://github.com/JIC-CSB/kilombo>

4.2.2 Kilobot Gym

The Kilobot Gym is a simulation framework designed to train an AI for the Kilobots (subsec 5.1) through reinforcement learning. It is based on OpenAI gym and Kilombo (subsec 4.2.1) ([Gebhardt and Neumann, 2018]).

It can be downloaded from GitHub through the website <https://github.com/gregorgebhardt/gym-kilobots>

4.2.3 Kilosim

The third simulation platform designed for the Kilobots (subsec 5.1) can also be used for other robotics swarms. The increase in efficiency stems from implementing a pseudo-physics engine, saving computational capacity in the process ([Ebert, 2022]).

As an open-source project, it can be downloaded from the website <https://github.com/jtebert/kilosim> and works as described in Algorithm (alg) 4.1.

Algorithm 4.1: Pseudo-code Simulation step in Kilosim

```
1 Init time t
2 for all robots do
3   | Run robot controllers (motors and battery state)
4   | Run user-implemented robot code
5 for all robots do
6   | Communicate between robots
7 for all robots do
8   | Compute next positions neglecting collisions
9 for all robots do
10  | Find collisions between robots and boundaries
11 for all robots do
12  | Move robots accounting for collisions
13  $t \leftarrow t + 1$ 
```

Chapter 5

Example projects

In the following, a selection of successful projects is listed.

5.1 Kilobots

Being the main research platform for all swarm robotics-related projects at Harvard University, a variety of experiments and simulation platforms have been documented and developed for the Kilobots. These coin-sized robots can collectively make binary decisions (Subsection (subsec) 3.3.3), recreate two-dimensional images (subsec 3.2.1), and solve inspection tasks (subsec 3.2). They are programmed in C ([Ebert, 2022], [Gebhardt and Neumann, 2018], [Hamann, 2018]).

Kilombo (subsec 4.2.1) and Kilosim (subsec 4.2.3) are used to simulate their movement while Kilobot Gym (subsec 4.2.2) is used to train an AI to solve inspection tasks. A picture of an individual robot can be found in Figure (fig) 5.1 and one of the Kilobots adapting a shape in fig 3.2.

Additionally, a graphical user interface called "KiloGUI" to upload code to the Kilobots is publicly available through <https://kilobotics.com/>. Options to buy Kilobots are on the same website.

Technical details taken from [Ebert, 2022] and [Gebhardt and Neumann, 2018]:

- Function: Collective decision-making making, two-dimensional picture reconstruction, and inspection tasks.
 - Actuators: 2 vibration motors connected to 3 equiangular rigid legs allowing the Kilobot to be able to walk around 1 cm/s. The direction of the movement can be
-

controlled by using only one motor.

- Sensors: Light sensing sensor and communication through an Infrared Communication (IR) transmitter.
- Communication: IR (subsec 6.2.2) transmission and reception possible within a 3-body radius. Each robot has a unique ID that was assigned during the initialization. The maximum communication rate is 10 messages/s, though this decreases at higher densities of robots.
- Limitations: limited battery life, noisy movement, and debugging can only be conducted by a single RGB Light Emitting Diode (LED) and a wired serial connection. Walking in a straight line is nearly impossible for them. Kilobots require a flat and low-friction surface to walk on due to movement throughout the slap-stick principle.

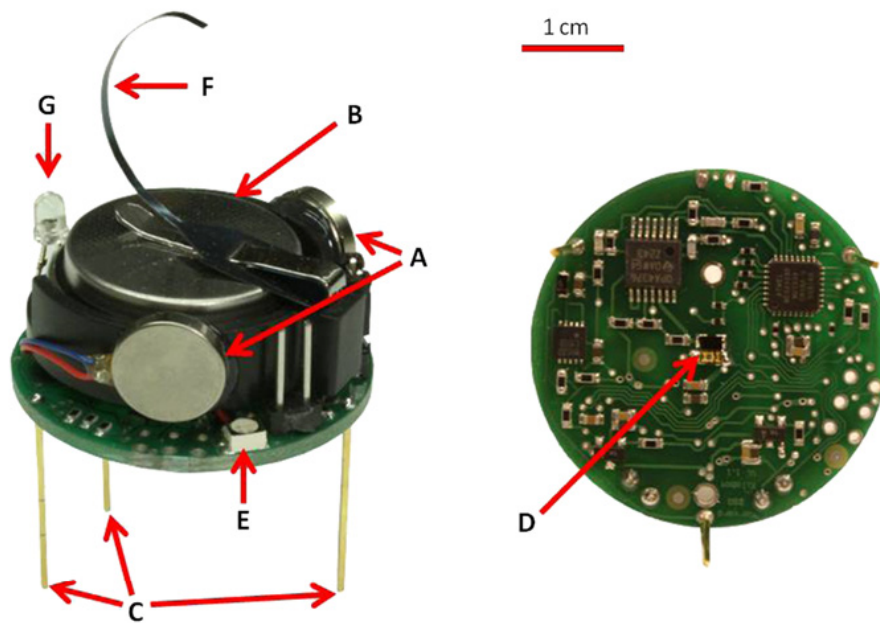


Figure 5.1: Kilobot

Single Kilobot robot. "Isometric (left) and bottom (right) views of a Kilobot. Some key features are: (A) vibration motors, (B) lithium-ion battery, (C) rigid supporting legs, (D) infrared transmitter/receiver, (E) three-color (RGB) LED, (F) charging tab, and (G) ambient light sensor". 1 cm line for scaling. Image and description taken from [Rubenstein et al., 2014]

5.2 TERMES

[Petersen et al., 2008] TERMES is a project to develop a swarm of robots capable of constructing three-dimensional objects out of special robotic boxes. Inspired by termites, it was developed at Harvard University from 2004 until 2008. These robots can move freely in any direction, including climbing up and down, collision avoidance, and picking up and attaching special passive building blocks ([Petersen et al., 2008]).

A more detailed explanation of the algorithm can be found in subsec 3.2.1 on Collective construction. A picture of a construction in progress in fig 3.1 and one of the robot itself in fig 5.2. A video of this swarm in action can be found here: <https://www.youtube.com/watch?v=MmQOEaXIANU>.

Technical details from [Werfel and Nagpal, 2018], [Petersen et al., 2008], and [Petersen and Nagpal, 2017]:

- Function: construction of 3-dimensional structures.
- Actuators: 3 unspecified actuators for movement.
- Sensors: IR, tilt sensors, and push buttons.
- Communication: receive a processed final image. At the start, a blueprint, a rule set, and navigation instructions via Bluetooth are sent. Later, information between the building blocks is done directly via touch (subsec 6.2.1). The mobile robots communicate implicitly (subsec 3.1.5).
- Limitations: the maximum climbing capacity is 1 building block.

5.3 S-bot and MarXbot

The S-bot has been further developed into both the foot-bots in the Swarmanoid project (subsec 5.4) and the MarXbot. The MarXbot added a lot of sensors (proximity sensors, a three-axis gyroscope, 3D accelerometer, and a distance scanner module) and improved the gripping mechanism that is now capable for use in uneven terrain. Thus, the marXbot is bigger and heavier than the S-bot ([Bonani et al., 2010]).

The S-bot is 12×15 cm big and features a Li-Ion battery with 10 Wh, allowing for two hours of usage. It boasts a PXA255 processor.

All technical details from the marXbot taken from [Christensen et al., 2008] and [Bonani et al., 2010], p.4188.

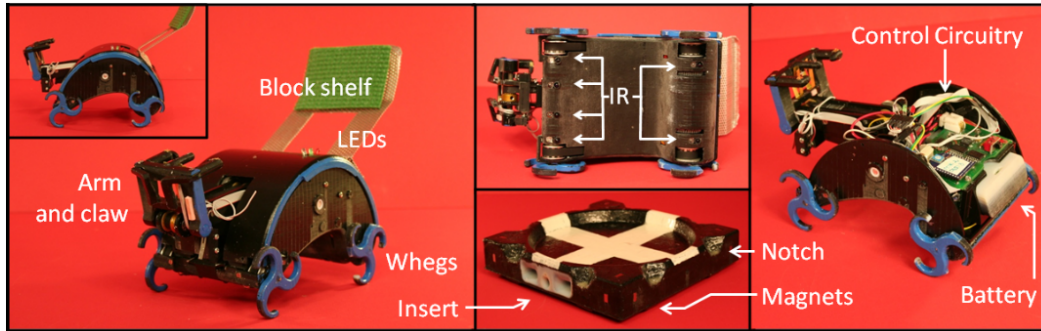


Figure 5.2: TERMES individual

TERMES individual robot. The gripper can be inserted in the "Insert" at the building blocks (lower middle picture). The movement in every direction without moving forward or backward is supported by the deeper part in the middle of the blocks. The robots know where the structure continues by following the black lines on the blocks ([Petersen et al., 2008])

- Function: attaching to each other; one of the members of Swarmanoid (subsec 5.4).
- Actuator: gripper.
- Sensors: omnica.
- Communication: Wi-Fi.
- Limitation: Can only fulfil its function.

5.4 Swarmanoid

The Swarmanoid-Project is based on the Swarm-bots (subsec 5.3) project ([Dorigo et al., 2019]). It was designed for usage in search and rescue missions and other complex tasks in human-made environments (subsec 3.2.2, subsec 28). Because of this, its name is a combination of the words "swarm" and "humanoid" ([Dorigo et al., 2013]). fig 5.3 and fig 5.4 show how the robots look. To test their algorithms, the developers of the Swarmanoid project have also created the simulation platform ARGoS (subsec 4.1.1) ([Pinciroli et al., 2012], [Pinciroli et al., 2011]).

As a heterogeneous group of robots, it could be argued that this project is not fully a product of swarm robotics according to subsec 3.1. Swarmanoid is, however, still included as an impressive example of autonomous collaboration of robots using a swarm

robotics approach.

Technical details from [Dorigo et al., 2019]:

- Function:
 - eye-bot: drone, can fly and find the desired object.
 - foot-bot: vehicle, can drive around and form a line as well as carry the hand-bot.
 - hand-bot: two-armed robot, that can climb slim objects with a roof on top through a ceiling attachment system.
- Actuators:
 - eye-bot: 8 brushless motors, a ceiling attachment device, and simple collision detection.
 - foot-bot: mobility and battery modules.
 - hand-bot: camera.
- Sensors:
 - eye-bot: camera (pan-tilt), distance sensor, 3d relative positioning sensor, 6 degrees of freedom inertial sensor, sonar, pressure, magnetometer.
 - foot-bot: camera, distance scanner, range and bearing module.
 - hand-bot: rope attachment and launching system, two grippers, and 2 fans.
- Communication:
 - eye-bot: led rings (visual), radio waves.
 - foot-bot: colours, radio waves.
 - hand-bot: radio waves.

5.5 InsBot

The InsBot was developed to test the interaction between a robot and a swarm of cockroaches (subsec 3.4.3). A picture can be found in fig 3.5. The design is based on a robot called "Alice".

It is 41x30x19 mm big and uses a 190-mAh Li-polymer battery, allowing it to run for 4 hours. A single robot costs 200 Euros. The battery can be fully recharged in 1 hour ([Caprari et al., 2005]).

All technical details from [Caprari et al., 2005]:



Figure 5.3: Swarmanoid: foot and hand-bot

3 foot-bots carrying a hand bot. The foot-bots look similar to the subsec 5.3. Taken from [Dorigo et al., 2019]

- Function: To prove that robots of that size can be built and manipulation of swarms of cockroaches.
- Actuators: watch motors.
- Sensors: IR sensors, light sensors, linear camera (for longer range perception), and temperature (to follow temperature gradient).
- Communication: local communication through IR (subsec 6.2.2), brightness and emittance. The latter one is used to distinguish other Insbots from obstacles. Manual software is uploaded through a bootloader using a serial port. There is a radio-based station receiver onboard used only for monitoring reasons. Each robot has a unique address.
- Limitations: Only works on cockroaches.

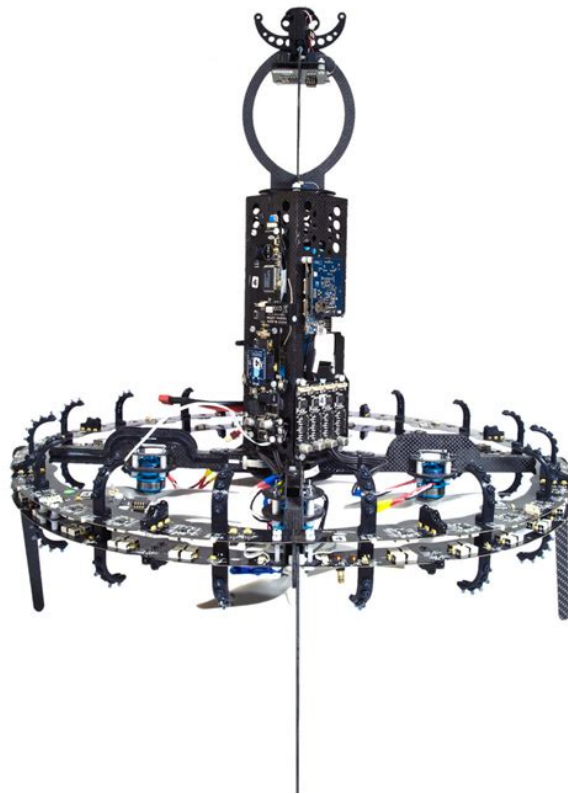


Figure 5.4: Swarmanoid: eye-bot

The eye-bot is a drone capable of Video and image recognition with several communication techniques. Taken from [Dorigo et al., 2019]

5.6 RiBot

The RiBot was developed to test the influence of a robot on a school of zebrafish. More information can be found in subsec 3.4.3 and a picture in fig 3.6.

Based on a 3D scan of a zebrafish, it mimics the shape and appearance of the animal as closely as possible, with the only difference being that the robot is about 1.5 times longer than zebrafish. The pattern was achieved through an outer layer of soft latex, polyurethane, and decals.

All technical details from [Bonnet et al., 2018]:

- Function: manipulation of a swarm of zebrafish.
- Actuators: 1 stepper motor for the tail. It swims about 3cm in height with the help of magnetic coupling to the blade driven by the motor underneath the tank.
- Sensors: IR-receiver.

- Communication: IR- communication (subsec 6.2.2) for RC5 signals for remote control of the mechanic fish.
- Limitations: was only able to influence 65% of fish. It only works on zebrafish.

5.7 ePucks

The ePucks are simple and sturdy, have a good variety of sensors, and are user-friendly, making them ideal for educational purposes. The goal of this robot is a practical implantation possibility for students looking to understand and program swarm robotics. An own communication module using radio waves (subsec 6.2.3) was developed for this purpose. The software can be downloaded from <https://e-puck.gctronic.com/>. Simulations can be run on both webots (Section (sec) 4.1.2 ([Cianci et al., 2007]) and ARGoS (sec 4.1.1).

All technical details are taken from the website <https://www.gctronic.com/e-puck.php>. The ePucks can be bought for 570 - 850€ through the same link. Its interchangeable 5 WH LiIon battery allows for about 3 hours of autonomy. It is programmed in C. This robot has received an upgrade. The ePuck2 is now commonly used.

1. Function: Educational robot. It can drive around and be custom-programmed.
 2. Actuators: the ones for movement were not explicitly named. A speaker is included.
 3. Sensors: 8 proximity and ambient light, 3D accelerometer, 3 omnidirectional microphones, VGA colour camera, and an IR receiver (subsec 6.2.2) for remote control are included.
 4. Communication: custom build module dsPIC at 60MHz (15MIPS) and a DSP core for signal processing, RS232 and Bluetooth connection.
-

Chapter 6

Hands On Approach: common issues

As discussed in the simulations chapter, many real-world difficulties cannot or only very hardly be modelled. Still, if these problems are not solved, the swarm will not work. Some creative solutions to these issues will be presented in this Chapter.

6.1 Battery

The short battery life is one of the main constraints in performing experiments with the swarms. Manually charging or exchanging batteries for a large number of robots might be tedious and consume a significant amount of time. A wired connection is less advantageous as it might lead to entanglement ([Arvin et al., 2017]). Thus, choosing the right battery system is a problem that every developer encounters while developing robotic swarms. Some have thought of new approaches to solve these problems.

6.1.1 Prolong the battery life

As there is only limited battery life, any member of the swarm can run out of battery at any given moment. One solution is to develop new, more efficient types of batteries. Development, however, takes time and money. Another approach is to include several batteries, just like the marXbot and the S-bots (Subsection (subsec) 5.3) did. However, as the overall weight and size of the robot is increased and a specialized circuit has to be developed, this approach does not significantly decrease production time or increase the operating time.

6.1.2 Automized exchange of batteries

TERMES (subsec 5.2) has proposed an alternative solution: the robots exchange their batteries on their own (subsec 3.2.1). Although the process was not explained in detail, it appears that two batteries and a navigation system are required to locate the battery station designed for this project. Additionally, the swarm behaviour has to be interrupted to exchange the batteries ([Arvin et al., 2017]). Therefore, it can be assumed that the development and usage of this feature presents a significant challenge, particularly for smaller and lighter robots with limited capacity for batteries and circuits.

6.1.3 Charging through the environment

The concept is similar to implicit communication. Why charge or change batteries explicitly if the environment in the form of the sun or the ground can do it instead? While solar power panels have been mounted to the robot Alice before, this is not a common solution due to the low effectiveness, limited availability and high weight of solar panels.

Another way to charge robots is through an electrical connection between the robot and a conductive ground. Although experiments have proven the feasibility, the charging capacity is significantly reduced when the connection faces disturbances in the form of dirt. Moreover, using this method restricts swarms specializing in collective construction (subsec 3.2.1) to specialized blocks, and other manipulations of the environment such as physical pheromone trails are not possible. Due to these limitations, the swarm is confined to a specific environment where the ideal circumstances are met ([Arvin et al., 2017]).

TERMES (subsec 5.2) has also included environmental charging in the form of the specialized building blocks charging each other once attached to the structure ([Petersen et al., 2008]).

6.1.4 Inductive charging pads

Another wireless charging option would be via inductive platforms in the working area through portable charging pads. Placing more pads in case of lower battery storage

capacity or the swarm increases in size has proven effective. Furthermore, it allows for an uninterrupted workflow. However, inductive charging is rather inefficient and only works over short distances. Additionally, the robot must have a special coil attached that might increase the weight. Still, the inductive coils weigh less than the solar panels ([Arvin et al., 2017]).

6.2 Communication

Communication is an essential topic when speaking of teamwork. Such is also the case in robot communication. There are many forms of communication, each with advantages and disadvantages depending on the situation.

6.2.1 Immediate proximity: Touch

TERMES (subsec 5.2), having already proposed to charge the building blocks through touch (subsec 6.1.3), also implemented the information transmission through touch. For this to work, the building blocks must share a stable, physical connection once attached to the structure ([Petersen et al., 2008]). As one can imagine, this solution is ambitious to implement in mobile robots that are constantly moving or susceptible to crashes from other moving individuals.

6.2.2 Close proximity: Infrared communication

Cheap, relatively easy, and small, Infrared communication (IR) is used to communicate with the environment and other members of the swarm and gather clues about its current operating state. Information is exchanged through variations of the frequency through patterns that can be decoded with protocols.

This technology is vulnerable to various types of noises including thermal noise, the signal itself, dark current, and ambient light. However, the sun and other weather phenomena such as fog have the most significant effect on the signal. In combination with its limited range of maximal 30 m indoors and 10 to 20 m outdoors, a swarm using exclusively this technology for communication should not be operated outside buildings ([Pauluzzi et al., 1992]). It is noteworthy that this paper is from 1992. Since then, Infrared Communication (IR) communication might have improved considerably.

Still, the fact remains that all swarms described in this paper use this technology and operate exclusively in lab environments.

6.2.3 Ranged communication: Radio Wave Communication

Communication via radio waves is the most straightforward way of explicit communication: large quantities of explicit and detailed information can be sent via a direct channel to a receiver. Due to the amount of information, it is also one of the more complicated means of communication. Still, this is the most common type of communication between mobile devices such as phones or radios. As such, many legal regulations are depending on the region of use.

Most frequencies are used for specific sub-types such as Bluetooth or Wi-Fi. As such, methods of interception are well known, increasing the security risk. Additionally, modules capable of radio wave communication on different frequencies tend to be significantly more expensive than, for instance, IR diodes. This method might therefore be most advantageous in heterogeneous swarms (subsec 3.1.2) or situations where global (subsec 3.1.4) or direct (subsec 3.1.5) communication is necessary. As such, this type of communication is heavily used in the Swarmanoid project (subsec 5.4).

Developing its own radio wave communication module for homogeneous swarms is also an option, as demonstrated by the ePucks (subsec 5.7).

6.3 Stray individuals

Be it due to a defect, a misunderstanding, or on purpose, what happens when a member of the swarm leaves its peers? In some cases, a robot will receive a message that it's supposed to return. In the other types of communication, it will simply not interact or influence the robots unless it naturally finds its way back to the swarm. The robot entering a 'panic' mode is also possible. Here, the robot will move in a random manner until it encounters its peers and integrates into the swarm again.

For algorithms such as BEECLUST (subsec 26), a robot's behaviour that differs from the norm may affect the swarm's behaviour. As a result, it should be taken into consideration when finding a solution.

Chapter 7

Summary

7.1 Summary

Swarm robotics is a relatively novel technology defined by its use of several relatively simple robots working together to reach a common goal. The main characteristics and advantages of swarm robotics are scalability, flexibility, and robustness, allowing for usage in different or changing environments (Subsection (subsec) 3.1) and a varying amount of swarm members (subsec 3.1.3). Usually, the swarms consist of similar individuals who only communicate if they are close to each other (subsecs 3.1.2, 3.1.4). The latter can be aided by using predominantly implicit communication techniques (subsec 3.1.5).

These robotic swarms are typically employed in the context of collective construction (subsec 3.2.1), collective manipulation and transportation of objects (subsec 3.2.3), search and rescue (subsec 3.2.2), and flocking and collective motion (subsec 3.2.4). Other areas include environmental exploration, both on and off our planet, agriculture, or inspection (subsec 3.2). To fulfil all these tasks, swarm robotics developing teams usually also have to solve questions like sorting (subsec 3.3.1), aggregation and dispersion (subsec 3.3.2), collective decision making (subsec 3.3.3), and optimization (subsec 3.3.4). Swarm robotics is closely related to Swarm Intelligence and shares many ideas and principles. The difference lies in the tangibility of the swarm (subsec 3.4.2). In this context, the interaction of swarm robotics with animals (subsec 3.4.3) and humans (subsecs 3.4.4, 3.4.5) was discussed. Difficulties in controlling a swarm and the phenomenon of emergence (subsec 3.5) were also highlighted.

Simulation platforms aid swarm robotics development by simplifying the testing process. Still, real-life robotic swarms are the goal. As such, examples for simulation platforms were provided in Chapter (chap) 4 and for robotic swarms in chap 5. Chapter 6 (chap 6) provides a hands-on discussion on practical solutions to recurring issues when designing robots meant to work in swarms. Finally, the last chapter (chap 7) provides a summary of the document and a short outlook on the further development of this technology.

7.2 Outlook

As hinted at in subsec 3.2, swarm robotics is a field with abundant potential for growth and different areas. Exploration of space, cleaning the ocean, improving warehouse transportation and all the things that the Kilobots can do - all of these are ongoing projects. Recently, research and development of new swarms seems stagnant but theses have been written about this topic at a growing rate.

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